

Chapter B8: Practical skills

Compliance with the requirements for practical work

It is compulsory that learners complete at least *eight* practical activities.

OCR has split the requirements from the Department for Education 'GCSE subject content and assessment objectives' – Appendix 4 into eight Practical Activity Groups or PAGs.

The Practical Activity Groups allow centres flexibility in their choice of activity. Whether centres use OCR suggested practicals or centre-substituted practicals, they must ensure completion of at least eight practical activities and each learner *must* have had the opportunity to use all of the apparatus and techniques described in the following table of this chapter.

The table illustrates the apparatus and techniques required for each PAG and an example practical that may be used to contribute to the PAG. It should be noted that some apparatus and techniques can be used in more than one PAG. *It is therefore important that teachers take care to ensure that learners do have the opportunity to use all of the required apparatus and techniques during the course with the activities chosen by the centre.*

Within the specification there are a number of practicals, indicated as, for example, *PAG1*, that are

described in the 'Assessable learning outcomes' column. These can count towards each PAG. We are expecting that centres will provide learners with opportunities to carry out a wide range of practical activities during the course. These can be the ones described in the specification or can be practicals that are devised by the centre. Activities can range from whole investigations to simple starters and plenaries.

It should be noted that the practicals described in the specification need to be covered in preparation for the questions in the written examinations that will assess practical skills. No less than 15% of the questions will assess practical skills. Learners also need to be prepared to answer questions using their knowledge and understanding of practical techniques and procedures in written papers.

Safety is an overriding requirement for all practical work. Centres are responsible for ensuring appropriate safety procedures are followed whenever their learners complete practical work.

Use and production of appropriate scientific diagrams to set up and record apparatus and procedures used in practical work is common to all science subjects and should be included wherever appropriate.

Revision of the requirements for practical work

OCR will review the practical activities detailed in Chapter B8 of this specification following any revision by the Secretary of State of the apparatus or techniques published specified in respect of the GCSE Biology B (Twenty First Century Science) qualification.

OCR will revise the practical activities if appropriate.

If any revision to the practical activities is made, OCR will produce an amended specification which will be published on the OCR website. OCR will then use the following methods to communicate the amendment to Centres: Notice to Centres sent to all Examinations Officers, e-alerts to Centres that have registered to teach the qualification and social media.

The following list includes opportunities for choice and use of appropriate laboratory apparatus for a

variety of experimental problem-solving and/or enquiry based activities.

Practical Activity Group (PAG)	Apparatus and techniques that the practical must use or cover	Example of a practical activity (a range of practicals are included in the specification and Centres can devise their own activity)*
1 Microscopy	Use of appropriate apparatus, techniques and magnification, including microscopes, to make observations of biological specimens and produce labelled scientific drawings ⁷	Investigate different magnification techniques to draw scientific diagrams from a number of biological specimens
	Use of appropriate apparatus to make and record a range of measurements accurately, including length, area, mass, time, temperature, volume of liquids and gases, and pH ¹	
2 Testing for biological molecules	Safe use of appropriate heating devices and techniques including use of a Bunsen burner and a water bath or electric heater ²	Testing foods for the presence of biological molecules in a range of foods
	Use of appropriate techniques and qualitative reagents to identify biological molecules and processes in more complex and problem-solving contexts including continuous sampling in an investigation	
3 Sampling techniques	Application of appropriate sampling techniques to investigate the distribution and abundance of organisms in an ecosystem via direct use in the field (to include: biotic and abiotic factors)	Investigation the differences in habitats using ecological sampling techniques
	Use of appropriate apparatus to make and record a range of measurements accurately, including length, area, mass, time, temperature, volume of liquids and gases, and pH ¹	
4 Rates of enzyme-controlled reactions	Safe use of appropriate heating devices and techniques including use of a Bunsen burner and a water bath or electric heater ²	Investigate the factors that can affect the rate of enzyme activity
	Use of appropriate apparatus and techniques for the observation and measurement of biological changes and/or processes ³	
	Measurement of rates of reaction by a variety of methods including production of gas, uptake of water and colour change of indicator ⁵	
	Use of appropriate apparatus to make and record a range of measurements accurately, including length, area, mass, time, temperature, volume of liquids and gases, and pH ¹	

Practical Activity Group (PAG)	Apparatus and techniques that the practical must use or cover	Example of a practical activity (a range of practicals are included in the specification and Centres can devise their own activity)*
5 Photosynthesis	Use of appropriate apparatus and techniques for the observation and measurement of biological changes and/or processes ³	Investigate the factors that can affect the rate of photosynthesis on <i>Cabomba</i>
	Safe and ethical use of living organisms (plants or animals) to measure physiological functions and responses to the environment ⁴	
	Measurement of rates of reaction by a variety of methods including production of gas, uptake of water and colour change of indicator ⁵	
	Safe use of appropriate heating devices and techniques including use of a Bunsen burner and a water bath or electric heater ²	
	Use of appropriate apparatus to make and record a range of measurements accurately, including length, area, mass, time, temperature, volume of liquids and gases, and pH ¹	
6 Physiology, responses respiration	Safe and ethical use of living organisms (plants or animals) to measure physiological functions and responses to the environment ⁴	Investigate the effect of exercise on pulse rate/ventilation rate and recovery
	Safe use of appropriate heating devices and techniques including use of a Bunsen burner and a water bath or electric heater ²	
7 Microbiological techniques	Use of appropriate apparatus and techniques for the observation and measurement of biological changes and/or processes ³	Investigate the effectiveness of antimicrobial agents on the growth of a bacterial lawn
	Use of appropriate apparatus, techniques and magnification, including microscopes, to make observations of biological specimens and produce labelled scientific drawings ⁷	
	Safe use of appropriate heating devices and techniques including use of a Bunsen burner and a water bath or electric heater ²	
	Use of appropriate apparatus to make and record a range of measurements accurately, including length, area, mass, time, temperature, volume of liquids and gases, and pH ¹	
8 Transport in and out of cells	Use of appropriate apparatus and techniques for the observation and measurement of biological changes and/or processes ³	Investigate the effect of different water potentials on the length and mass of potato chips
	Use of appropriate apparatus to make and record a range of measurements accurately, including length, area, mass, time, temperature, volume of liquids and gases, and pH ¹	

* Centres are free to substitute alternative practical activities that also cover the apparatus and techniques from DfE: Biology, chemistry and physics GCSE subject content, July 2015 Appendix 4.

^{1, 2, 3, 4, 5, 7} These apparatus and techniques may be covered in any of the groups indicated. Numbers correspond to those used in DfE: Biology, chemistry and physics GCSE subject content, July 2015 Appendix 4.

Choice of activity

Centres can include additional apparatus and techniques within an activity beyond those listed as the minimum in the above tables. Learners *must* complete a *minimum* of eight practicals covering all the apparatus and techniques listed.

The apparatus and techniques can be covered:

- (i) by using OCR suggested activities (provided as resources)
- (ii) through activities devised by the Centre.

Centres can receive guidance on the suitability of their own practical activities through our

free practical activity consultancy service.

E-mail: ScienceGCSE@ocr.org.uk

Where Centres devise their own practical activities to cover the apparatus and techniques listed above, the practical must cover all the requirements and be of a level of demand appropriate for GCSE. Each set of apparatus and techniques described in the middle column can be covered by more than one Centre devised practical activity e.g. “measurement of rates of reaction by a variety of methods including production of gas, uptake of water and colour change of indicator” could be split into two or more activities (rather than one).

NEA Centre Declaration Form: Practical Science Statement

Centres must provide a written **practical science statement** confirming that reasonable opportunities have been provided to all learners being submitted for entry within that year’s set of assessments to undertake at least **eight** practical activities.

The practical science statement is contained within the NEA Centre Declaration Form which can be found on the OCR website at www.ocr.org.uk/formsfinder. By signing the form, the centre is confirming that they have taken reasonable steps to secure that each learner:

- a) has completed the practical activities set by OCR as detailed in Chapter B8
- b) has made a contemporaneous record of:
 - (i) the work which the learner has undertaken during those practical activities, and
 - (ii) the knowledge, skills and understanding which that learner has derived from those practical activities.

Centres should retain records confirming points (a) to (b) above as they may be requested as part of the JCQ inspection process. Centres must provide practical science opportunities for their learners. This does not go so far as to oblige centres to ensure that all of their learners take part in all of the practical science opportunities. There is always a risk that an individual learner may miss the arranged practical science work, for example because of illness. It could be costly for the centre to run additional practical science opportunities for the learner.

However, the opportunities to take part in the specified range of practical work must be given to all learners. Learners who do not take up the full range of opportunities may be disadvantaged as there will be questions on practical science in the GCSE (9–1) Biology B (Twenty First Century Science) assessment. Please see the JCQ publication *Instructions for conducting non-examination assessments* for further information.

Any failure by a centre to provide a practical science statement to OCR in a timely manner (by means of an NEA Centre Declaration Form) will be treated as malpractice and/or maladministration [under General Condition A8 (*Malpractice and maladministration*)].

2d. Prior knowledge, learning and progression

- Learners in England who are beginning a GCSE (9–1) course are likely to have followed a Key Stage 3 programme of study.
 - GCSEs (9–1) are qualifications that enable learners to progress to further qualifications, either Vocational or General.
 - There are no prior qualifications required in order for learners to enter for a GCSE (9–1) in Biology B (Twenty First Century Science).
- There are a number of Science specifications at OCR.
Find out more at www.ocr.org.uk